

UAB

ICD-10 Clinical Codification

NLP-I Subject Q2 2025-2026



The Clinical Codification Problem

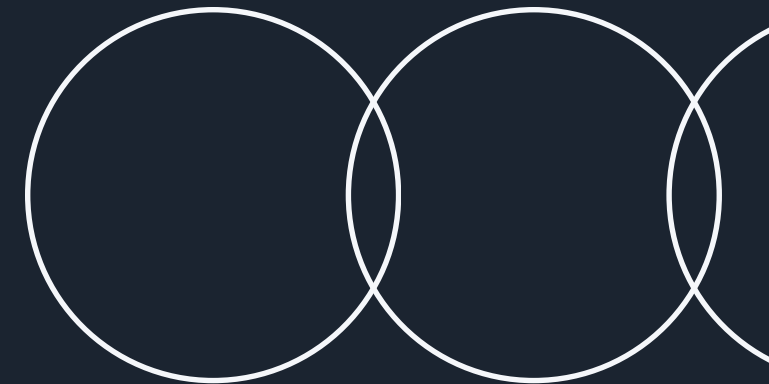
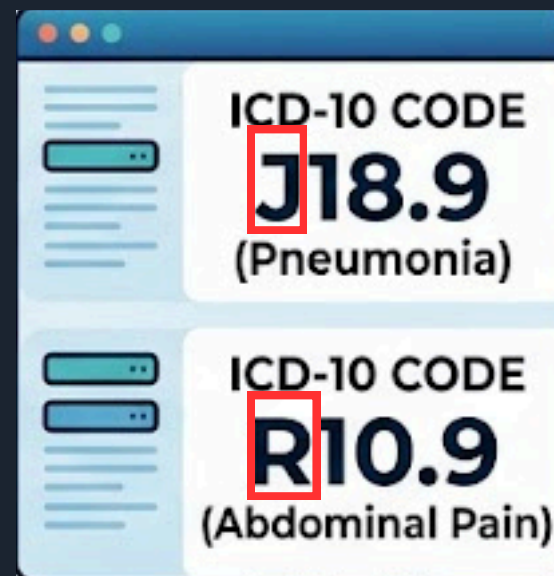
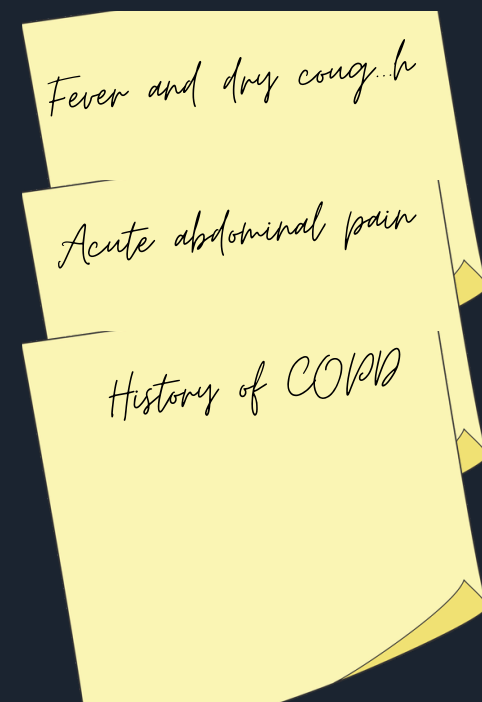
Understanding ICD-10 Importance

Classification

Clinical codification involves mapping medical records to standardized alphanumeric codes.

Challenge

Predict the ICD-10 code, based on a short clinical literal.



Data Analysis

Key Issues and metrics in Clinical Codification Process

Domain

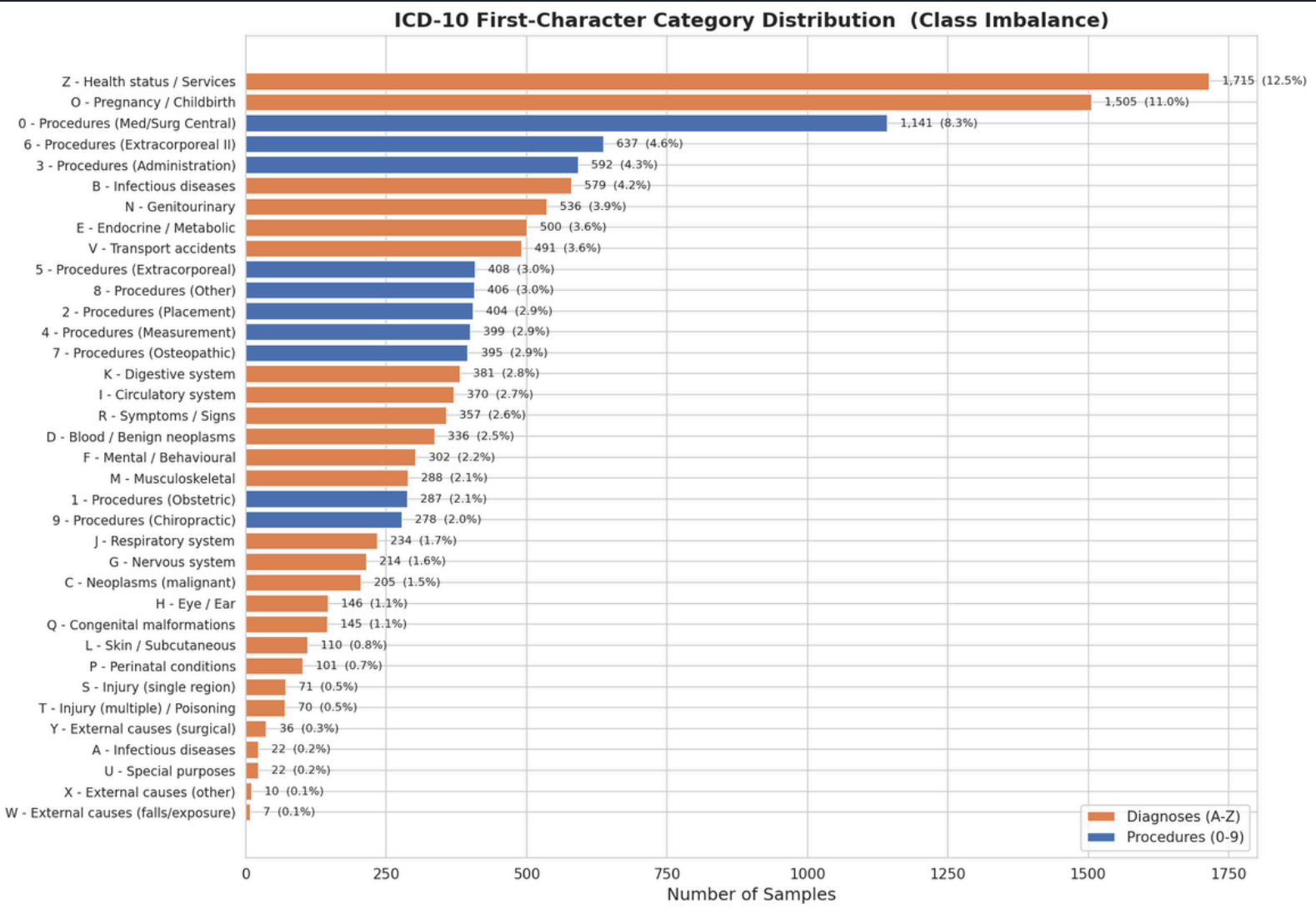
The ICD-10 code uses letters and numbers. Letters (A-Z) diagnosis and numbers (0-9) medical, surgical, and obstetrical procedures.

Corpus Statistics

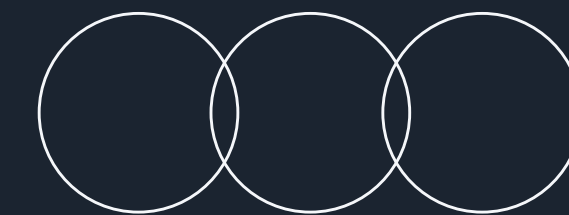
13.700 literals with a mean character length: 17 characters and a mean word count: 2 words. Very short literals.

Class Imbalance

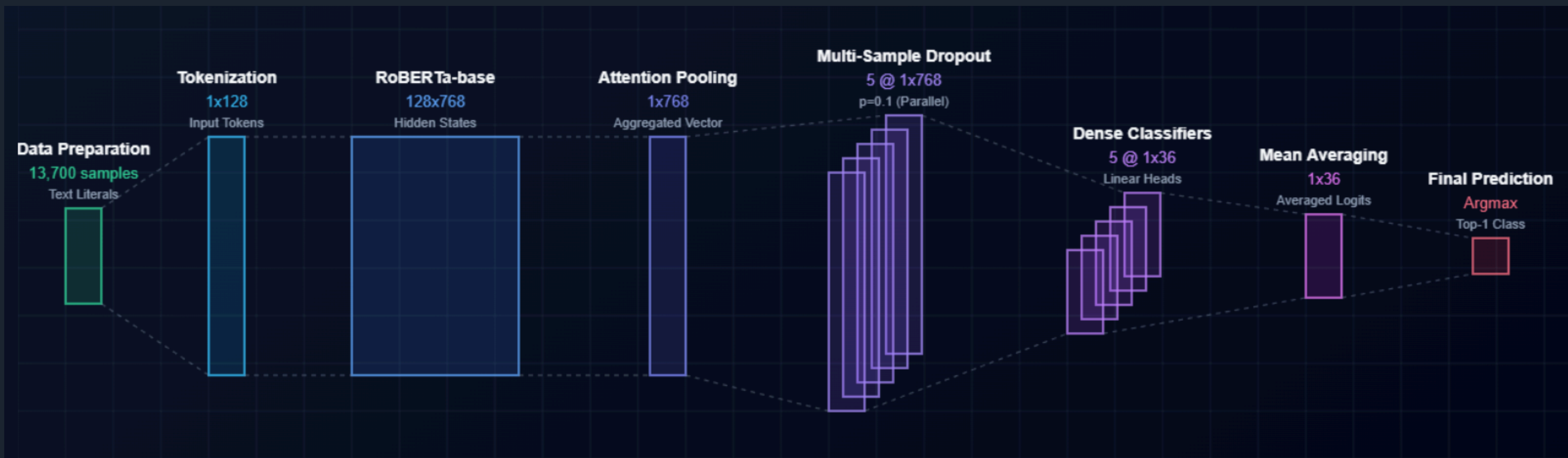
The dataset suffers from severe class imbalance, where classes Z, O, and 0 have more than 1000 samples each one. While there are some clases with less than 30 samples



Architecture Overview

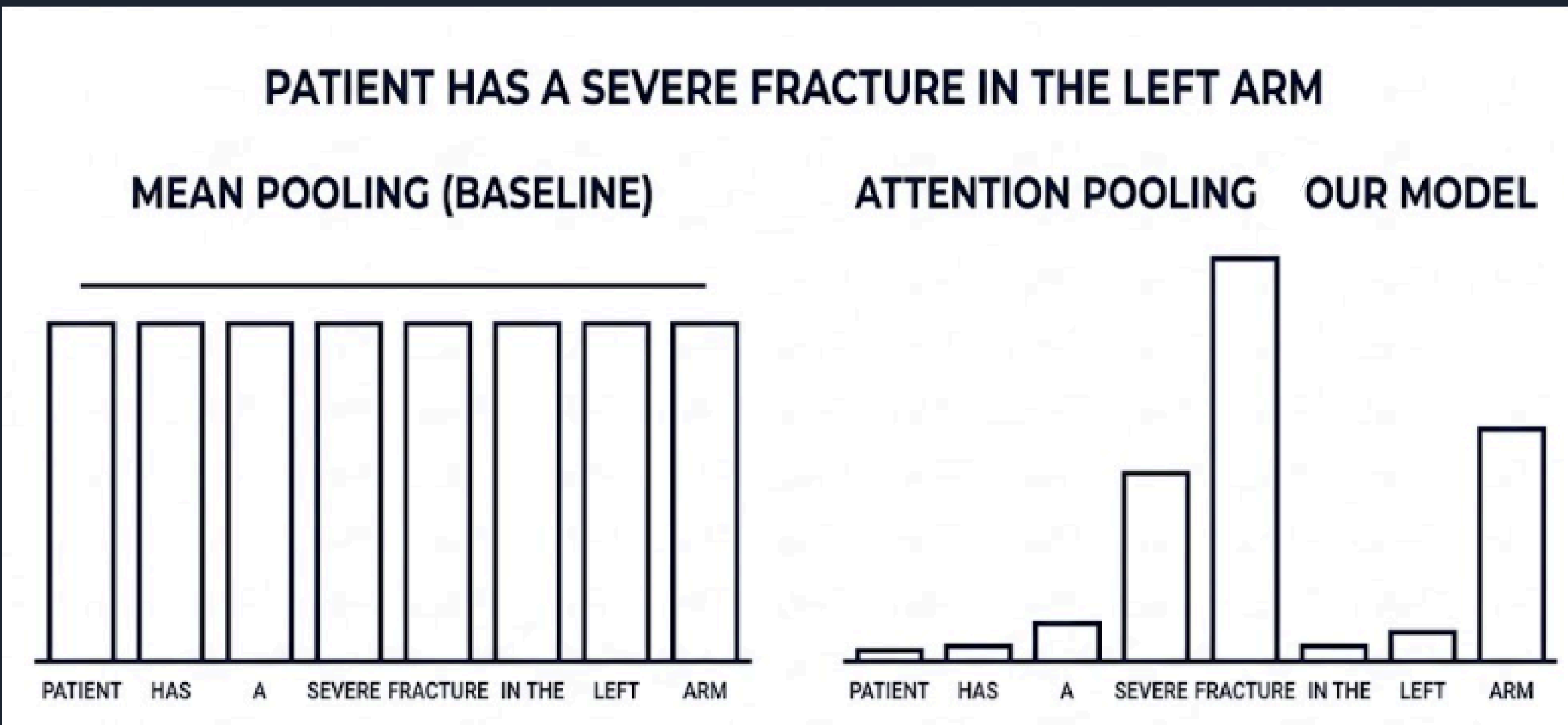
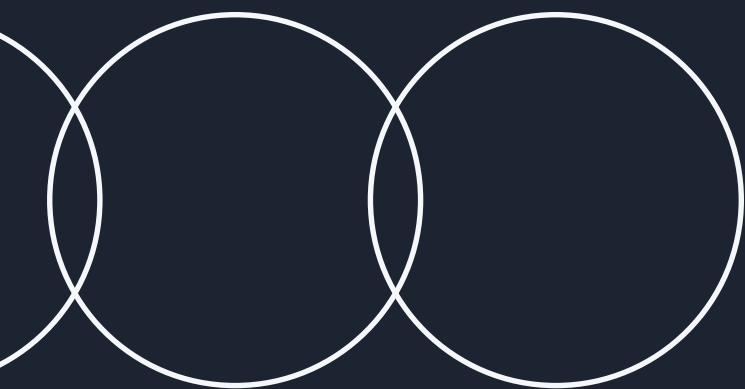


Key Components and Main Differences



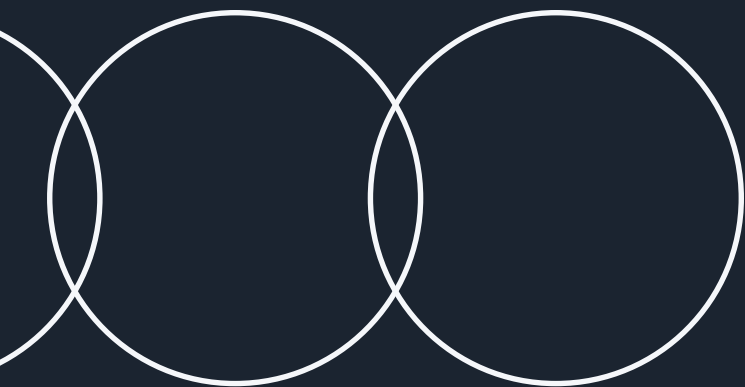
Custom Attention Pooling

Enhancing model performance
through tailored techniques

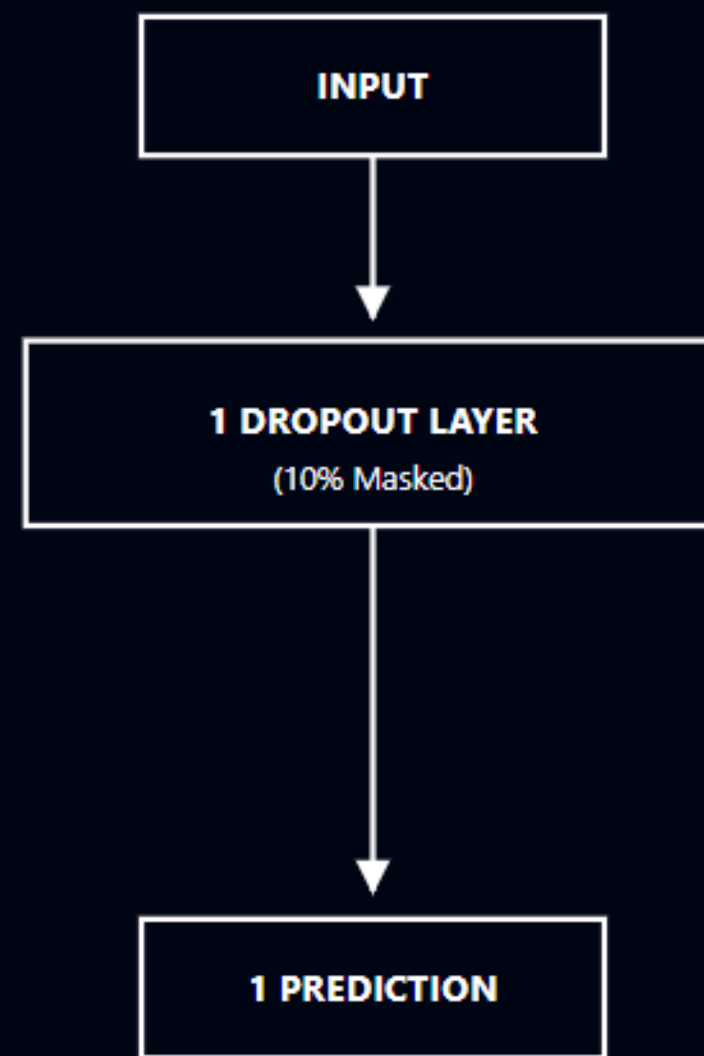


Multisample Dropout

Enhancing model robustness through dropout strategies

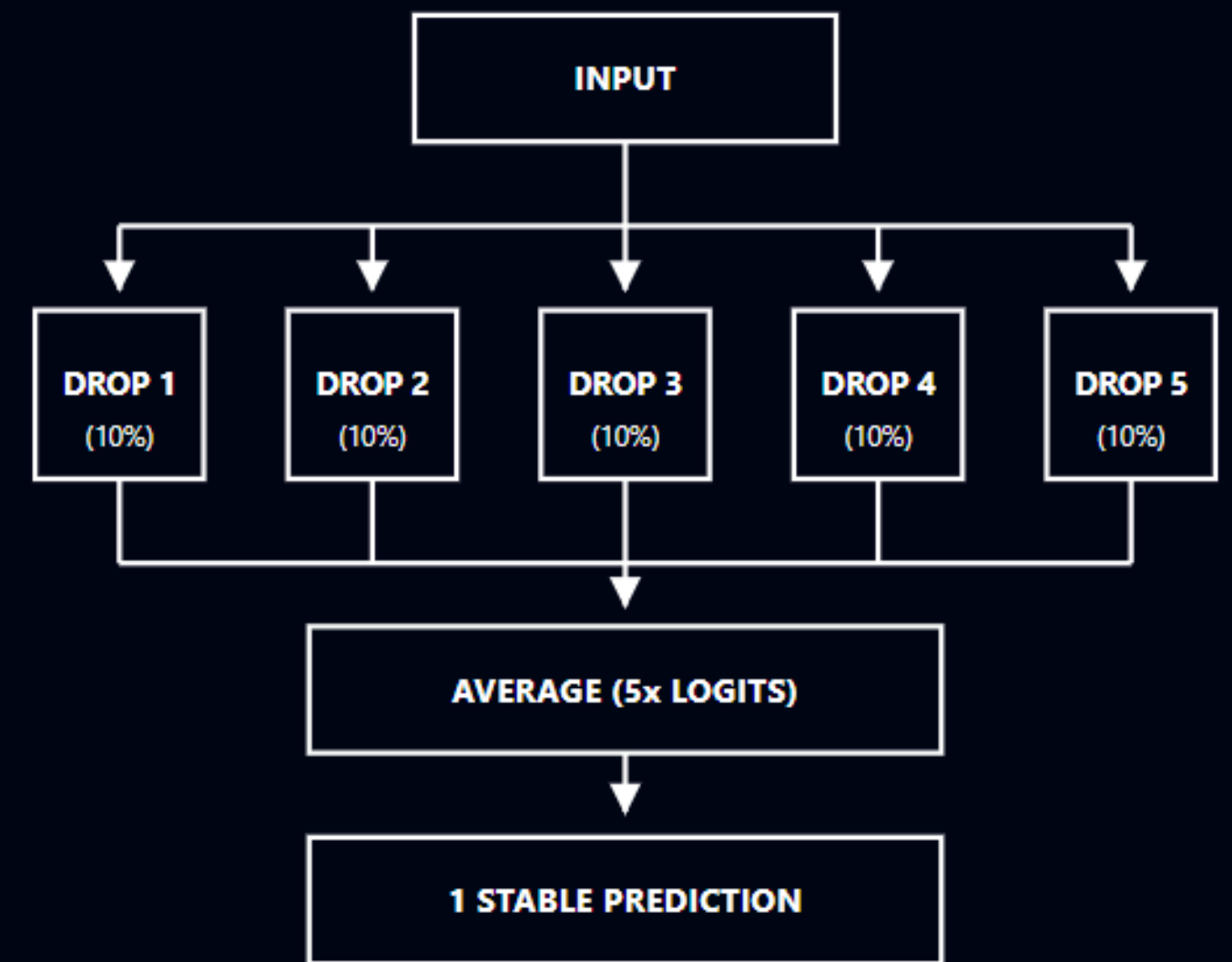


STANDARD DROPOUT



Single random mask.
High variance, prone to overfitting.

MULTI-SAMPLE DROPOUT



Ensemble of 5 parallel masks.
Highly stable, protects rare classes.

Data Preparation

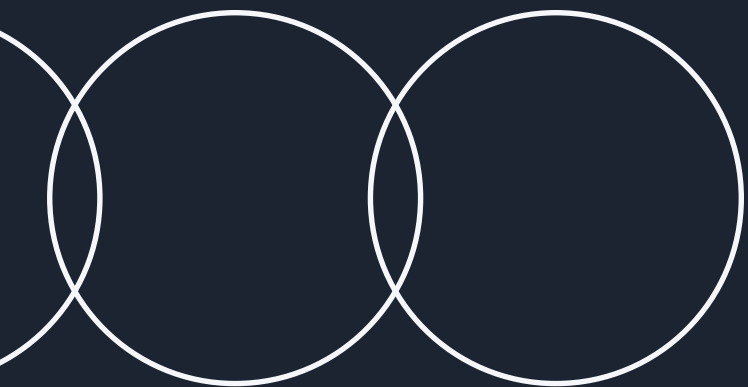
Refining the data to obtain an stable training

Filtering

Our data preparation phase focused entirely on filtering the records down to valid alphanumeric ICD-10 chapters, followed by a normalization layer to strip anomalies while preserving data relevance.

Stratified Split Optimization

We preserved the absolute empirical purity by training exclusively on real clinical records with a 80/20 stratified split (preserving class ratios).



Training Engineering

Optimizing the Deep Learning Model Effectively

Full-capacity fine-tuning

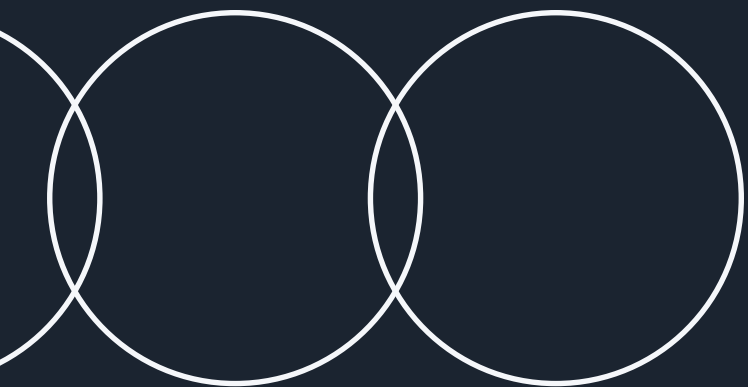
We used the 126 million parameters for the training

Baseline-Aligned Loss Function

We utilized an unweighted CrossEntropyLoss to keep the architectural comparison to the baseline mathematically identical

Training optimization

We applied various methods to prevent running out of memory



Results Overview

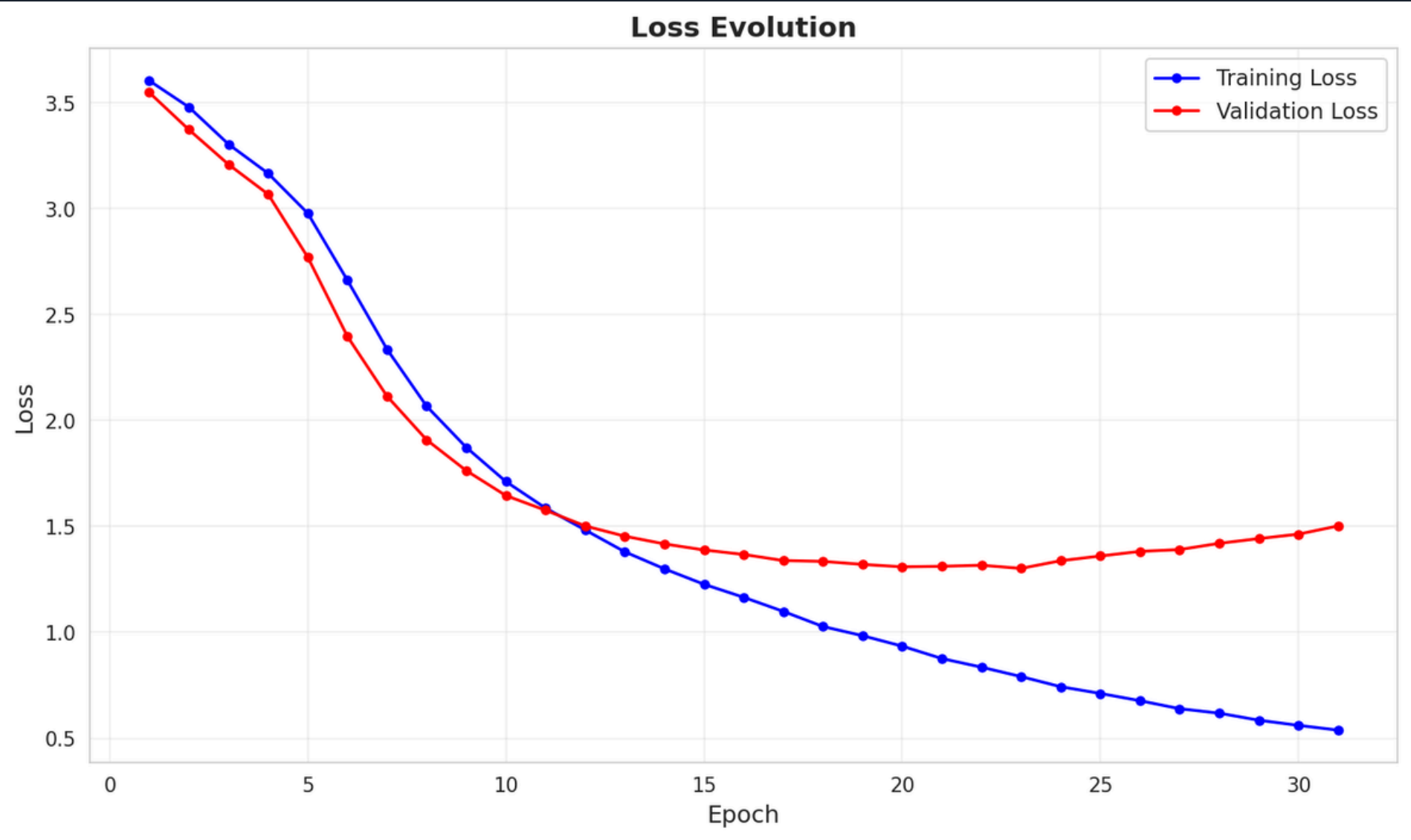
Metric	TF-IDF + Logistic Regression	RoBERTa + Attention Pooling	Delta (Δ)
Accuracy	52.99%	57.01%	+4.02%
Macro- F_1	44.49%	49.42%	+4.93%
Weighted- F_1	51.04%	55.07%	+4.03%

Baseline Pipeline (TF-IDF + Logistic Regression)

The baseline model using CLS pooling achieved an accuracy of 56.5% and a macroF1 score of 44.49%, indicating limited performance.

Baseline RoBERTa + Attention Pooling

The mean pooling baseline exhibited an accuracy of 57.0% and a macroF1 score of 49.42, slightly improving upon CLS pooling results.



Accuracy Limitations

Understanding the complexity of performance metrics in AI

Class Distribution

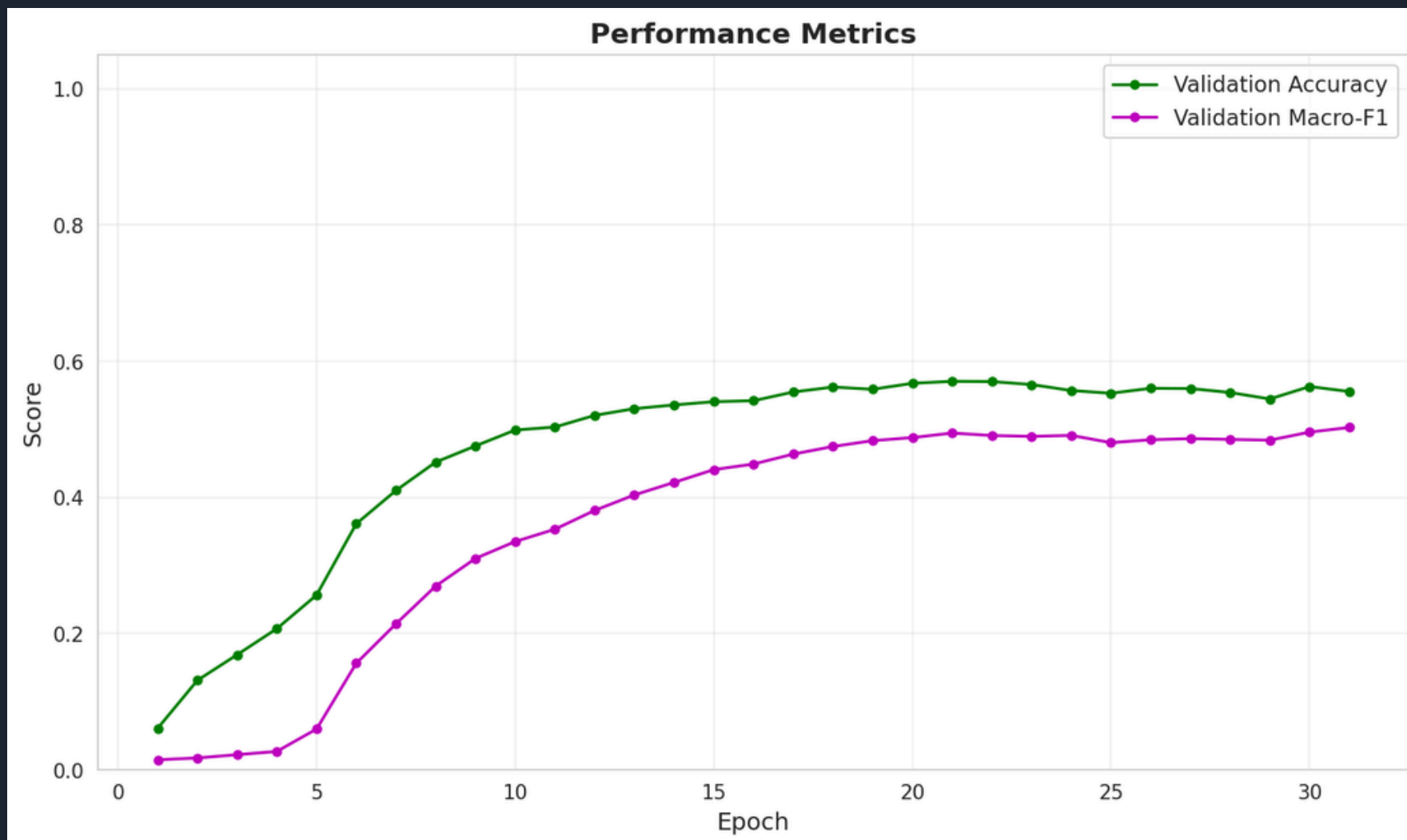
Classes Z, O, and 0 dominate the dataset, constituting over **30%** of the training samples, leading to potential biases in predictions.

Simple Model Baseline

A naive model predicting only these three classes achieves **60% accuracy**, which can mislead stakeholders about the model's overall effectiveness.

Importance of MacroF1

The MacroF1 score evaluates all **36 classes** equally, highlighting the model's performance on rare diagnoses and providing a more nuanced accuracy assessment.



Errors and Limitations

Understanding challenges in clinical codification accuracy and reliability

Ambiguity Issues

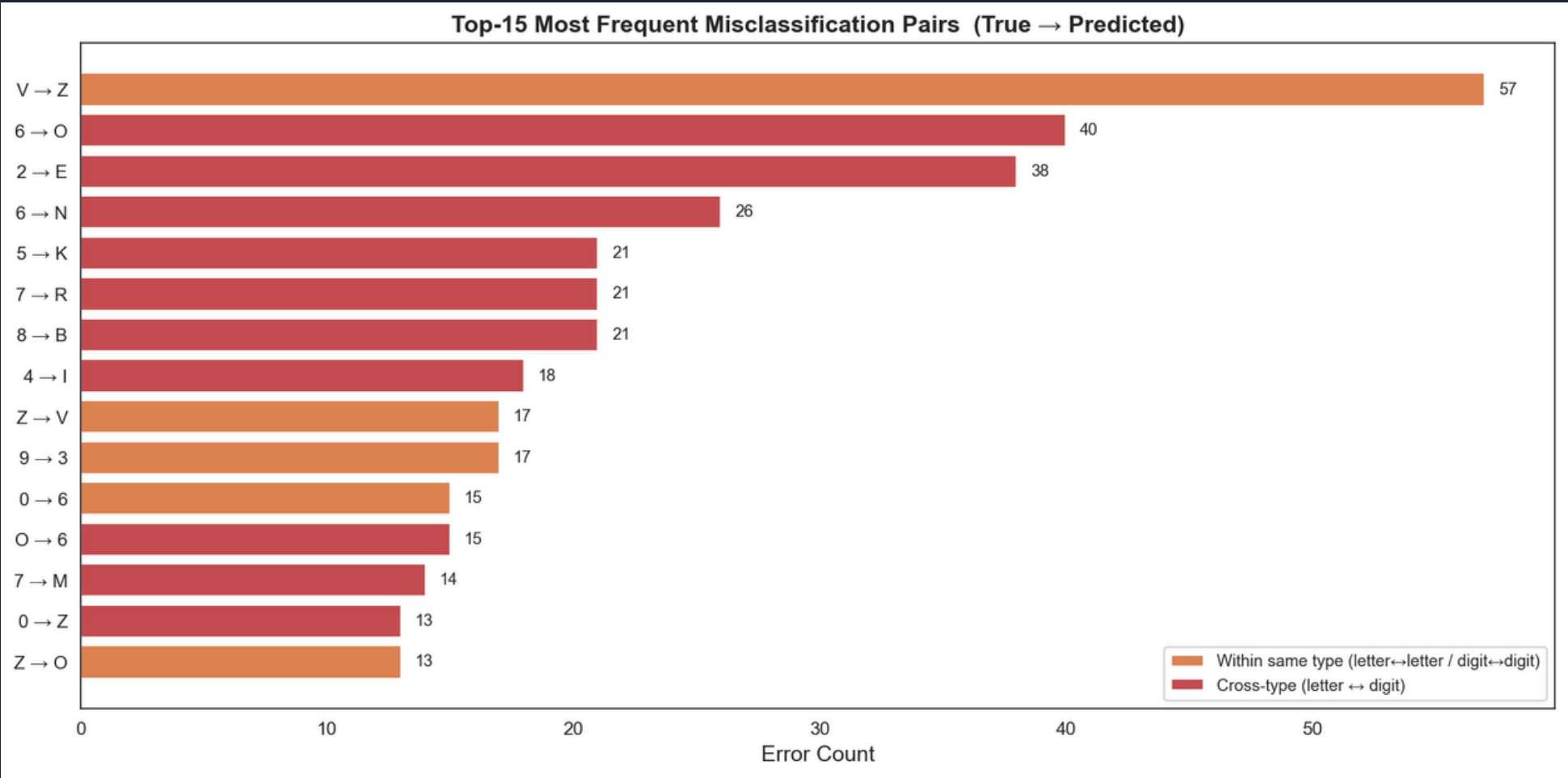
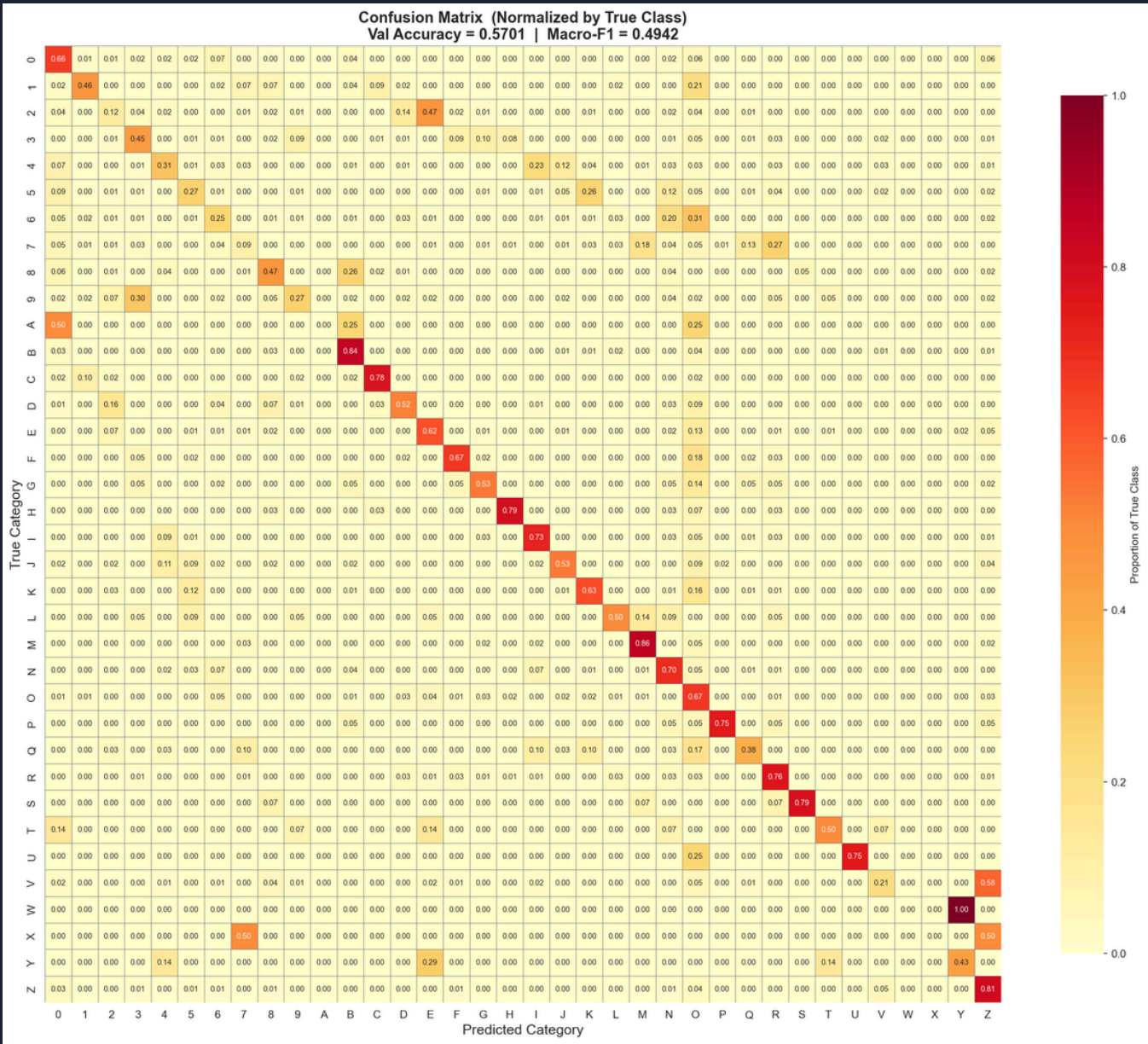
Ultrashort literals can lead to **misinterpretations**, causing incorrect mappings to ICD-10 codes due to lack of context.

Boundary Confusion

Distinctions between endocrine and general symptoms may be **blurred**, leading to potential misclassification of clinical notes.

Domain Mismatch

The disparity between lengthy dictionary entries and **concise clinical notes** creates challenges in accurate codification and classification.



Ethics and Conclusions

Addressing biases and future directions for research

Data Bias

The training data reflects **common hospital cases**, leading to potential biases in diagnosing less frequent conditions in the Catalan healthcare system.

Risk Assessment

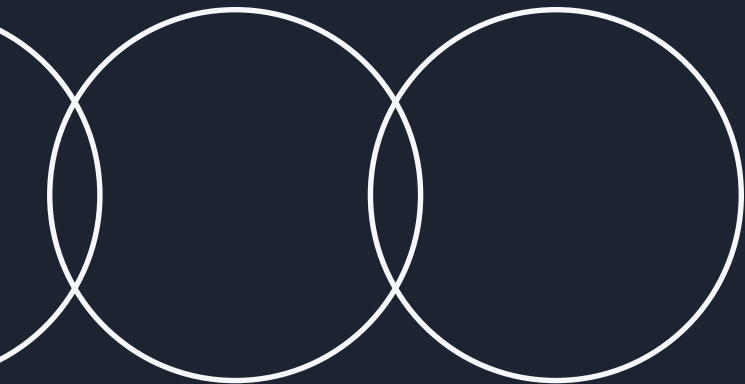
There is a **significant risk** of underperformance on rare conditions, which may negatively impact underrepresented demographics in healthcare.

Key Engineering takeaways

Keeping the validation split clean before augmentation (without this, all metrics are inflated and meaningless)

Switching early stopping from accuracy to Macro-F1 (without this, the model stops when majority classes converge and never learns rare ones)

Multi-sample dropout + class-weighted loss (these together are responsible for the MacroF1 jump)



What are your questions?

Thank you for your attention and engagement!

